

# 6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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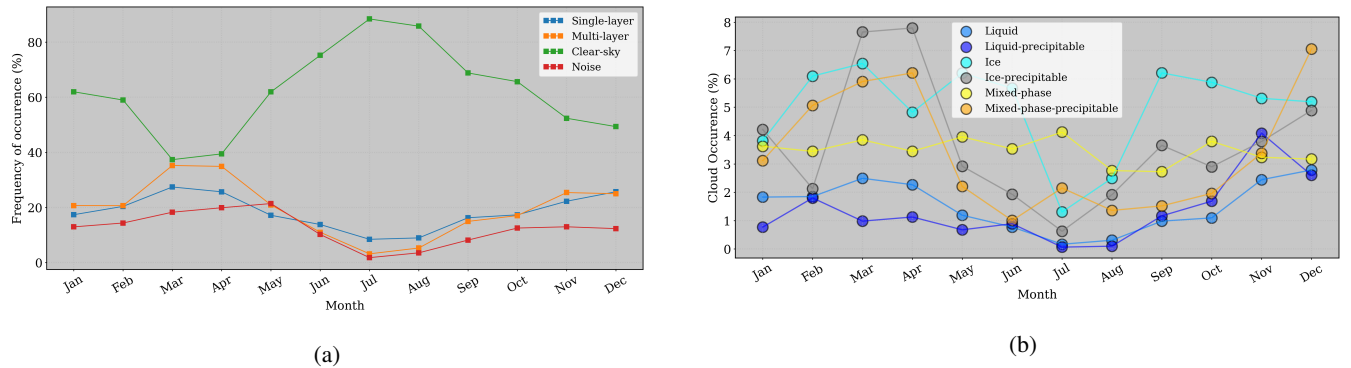
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## 1 Results and discussion

### 1.1 Annual variability and daily cycle of clouds

Since the cloud classification by hydrometers grouping showed a more comprehensive classification, this algorithm is employed in the following analysis. This section aims to analyze the inter-annual and daily cycle variability of cloud frequency of occurrence at the Granada station. Therefore, left panel of Figure 1 shows the monthly frequency of occurrence of different sky conditions, categorized into single-layer clouds, multi-layer clouds, clear sky, and noise. The data reveal a distinct seasonal pattern in cloud cover and clear sky occurrences. Clear sky conditions (green line) are predominant all year long, being most frequent in the summer months (80%), peaking in July and August, with a significant decline in spring (40%), reaching the lowest point in March and April. This pattern indicates a higher prevalence of clear skies during the warmer months and increased cloudiness during the colder months. Single-layer clouds (blue line) exhibit a relatively weak seasonal pattern throughout the year, with a slight peak in spring (35%) and a decrease in the summer months (10%), suggesting a moderate variation in single-layer cloud occurrences. Multi-layer clouds (orange line) show a similar pattern, with a peak in March and April and a reduction during the summer, indicating a seasonal preference for multi-layer clouds in the spring. The noise category (red line) maintains a low and stable occurrence throughout the year, with minor fluctuations. Overall, the data highlight clear seasonal variations in sky conditions, with clear skies dominating in summer and increased cloudiness, both single and multi-layer, in the cooler months. Since this station exhibit a quit low cloud occurrence, a need to handle multilayer clouds is revealed to fill gaps of cloud properties at regions such as the western Mediterranean. Particularly, these multi-layer are problematic due to inaccurate LWP partitioning, leading a misclassification of liquid pixels inside clouds, as already discussed by Tatiana Nomokonova et al.. Therefore, the following analyses are focus only on single-layer clouds, seeing their physical properties can be retrieved more accurately.

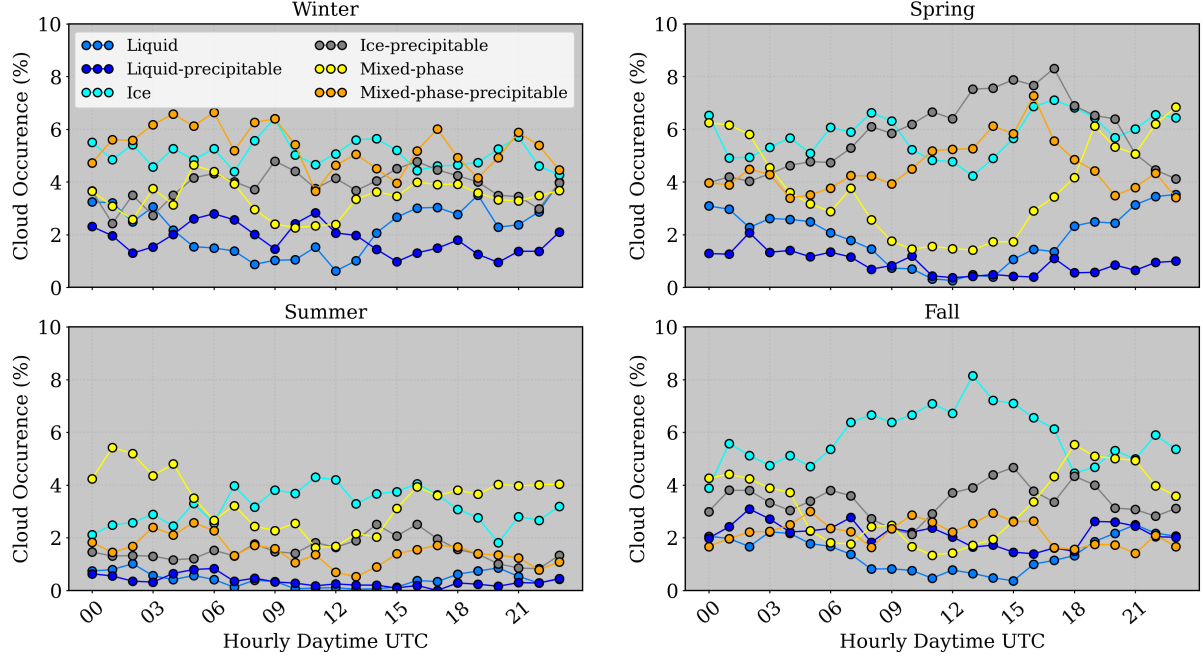
Monthly frequency of occurrence is depicted in the right panel of Figure 1. Mixed-phase clouds (yellow) exhibit relatively consistent occurrences with slight fluctuations around 3% and 4%. Liquid-precipitable (dark blue) has a modest variation with zero occurrence in summer, followed by an increase during the transition to winter, reaching a maximum of 4% in November, suggesting increased precipitation during fall. Liquid clouds (blue) are more variable, showing a percentage of 3% in March and December, against zero in summer. Ice clouds (cyan) drops sharply in July and maintain almost constant levels around 5% and 7% throughout the rest of the year, indicating hotter temperatures that inhibit ice cloud formation during summer. Ice-precipitable clouds (gray) have a notable peaks of 8% in spring, followed by a decrease in summer (1%) and an increase in fall, until reach a local maximum of 5% during winter. It suggest a seasonal variations in ice-related precipitation with maximum in spring, unlike the case of water-related precipitation, which showed a maximum in November. Mixed-phase-precipitable clouds (orange) peak in April (6%) and December (7%), indicating higher chances of mixed-phase precipitation towards spring and the end of the year. Overall, ice, ice-precipitable, and mixed-phase clouds have the strongest seasonal pattern, indicating a higher sensibility of ice due to thermodynamic conditions than liquid droplets. Moreover, the precipitation in this region is intrinsic related to ice and mixed phase clouds i.e. with low temperatures.



**Figure 1.** Inter-annual single layer cloud frequency of occurrence. The frequency of each cloud type is indicated in the legend.

An important mechanism associated with temperature decreases and responsible for extreme rainfall in Eastern Spain is the occurrence of cut-off lows (COLs). COLs account for nearly all rain events in Valencia, Spain (Ferreira), and are likely to affect the Granada station as well. This influence is attributed to the easterly wind component of COLs formed in the warm Mediterranean (Porcù et al.), which can reach the station, bringing humidity and cold air masses that favor cloud formation. In Valencia, Ferreira reported the rainiest days occurring in winter, with extreme rainfall events in spring and fall, aligning with our observations. During the summer, although almost non-precipitating clouds were observed at our station, temperatures were higher. Additionally, the Iberian Peninsula experiences a large low-pressure area over northern Africa, with high-pressure systems over the Atlantic Ocean and eastern and central Europe. This results in easterly winds over Andalusia, transporting moisture from the Mediterranean Sea (F. J. Bello-Millán et al.). This combination of high temperatures and humidity can create an environment conducive to storm formation. In this context, the few clouds observed in summer are likely related to convectively formed cumulonimbus clouds. This will be more evident when study their physical properties in further sections.

Next, the daily cycle of cloud occurrence should uncover the influence of temperature around mid-day, when the solar radiance is more intense. Although we already discuss the influence of synoptic and local phenomena in cloud formation, convection clouds should appear in the diurnal cycle. From Figure 2, its clear that there is no convection for any type of cloud during summer, fall and winter. Despite that fall shows an increase of ice clouds around mid-day, they are probably cirrus clouds, since their median base height and thickness are 8.5 km and 1.5 km, respectively (see Figure 3).



**Figure 2.** Inter-annual single layer cloud frequency of occurrence. The frequency of each cloud type is indicated in the legend.

However, spring shows a possible presence of convective clouds, with a slight peak of occurrence for Ice-precipitable and Mixed-phase-precipitable clouds between 09:00 and 21:00 UTC. Different from non-precipitating clouds which reached a minimum around mid-day, these clouds doubles the occurrence from 4% at 00:00 to 8% at 16:00-17:00 UTC. These observations are consistent with the findings of Shahi et al., where convection-permitting models of the Iberian Peninsula (IP) captured high values of mean precipitation during spring. Additionally, Alexandre Rios-Entenza et al. revealed that specific synoptic conditions, such as a reduction in the strength of the zonal circulation, can favor local and mesoscale convection during spring in the western region of the IP. This analysis confirms that the maximum cloud and rain occurrence observed during peak sunlight hours may be strongly associated with local convection.

## 1.2 Seasonal patterns in cloud base, top, and thickness

Seasonal evolution of cloud base height, top height and thickness heights are displayed in Figure 3. This figure shows that precipitating clouds exhibit a significantly lower cloud base height (top panel) compared to non-precipitating clouds, as well

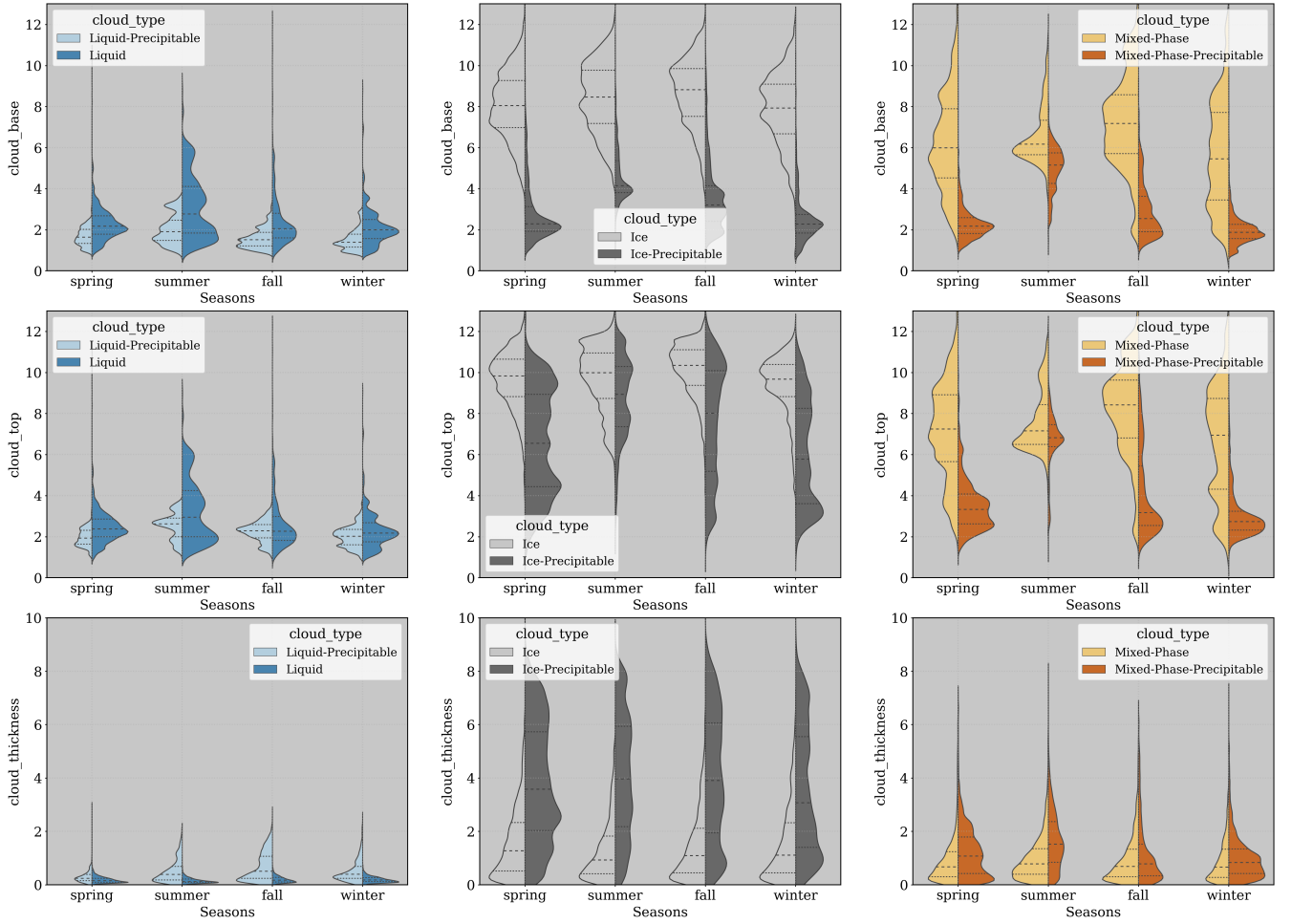
as a narrower distribution. Regarding non-precipitating ice clouds, they possess the highest cloud base heights, with median values around 8 km. These high ice clouds with small ice water path (Figure 9) can be associated to cirrus clouds. Next, non-precipitating mixed-phase clouds are the second highest cloud type we found, which show considerable variability in height except during summer, when values stabilize at 6 km. Differently, cloud base is high in fall, where the 50th percentile reach a maximum of 7 km of height, along with high variability. On the other hand, non-precipitating liquid clouds showed higher cloud base variability during summer, achieving significantly different heights compared with other seasons. Overall, in summer, liquid clouds, precipitating mixed-phase clouds, and ice precipitating clouds demonstrate elevated cloud base heights, attributed to the relatively high temperatures and low humidity at this season. Specifically, liquid clouds display the highest cloud base variability during summer, with the 50th and 75th percentiles around 4 km and 6 km, respectively. These clouds can be related to stratocumulus and stratus clouds, which were dis-coupled from cumulonimbus at final stage.

As observed for the cloud base, precipitating clouds have a significantly lower median cloud top height (middle panel of Figure 3) compared to non-precipitating clouds and exhibit a narrower cloud top distribution, except in the case of ice clouds. Non-precipitating ice clouds maintain a similar distribution across seasons, highlighting another feature that suggest the presence of cirrus clouds, while ice-precipitable clouds demonstrate highly variable cloud tops. Given their well-defined cloud bases, these clouds exhibit greater cloud thickness, indicative of storm activity at the station. The cloud tops for liquid clouds mirror the variability of their cloud bases, particularly during summer. These clouds exhibit a well-defined cloud thickness, with purely liquid clouds having a thickness below 1 km and liquid-precipitable clouds reaching up to 1 km in fall, as indicated by the 75th percentile. This aligns with the observed peak occurrence of these clouds in this season. Mixed-phase clouds also follow the behavior of their cloud bases and display a well-defined cloud thickness. Generally, non-precipitating mixed-phase clouds have a thickness around 500 m, increasing to 1 km for precipitating mixed-phase clouds, with slightly higher values in summer. For non-precipitating mixed-phase clouds, median cloud tops vary around 7 km during spring, summer, and winter, but reach 8.5 km in fall. Precipitating mixed-phase clouds have cloud tops around 3 km in spring, fall, and winter, peaking at 7 km in summer.

Overall, ice-precipitable clouds exhibit the greatest cloud thickness, followed by ice, mixed-phase-precipitable, mixed-phase, liquid-precipitable, and liquid clouds. It is important to note that, despite Cloudnet post-processed data being corrected for liquid attenuation, cloud tops might be underestimated due to high attenuation in some cases, reducing radar sensitivity.

### 1.3 Annual and seasonal cycle of liquid water content

The liquid water content in Figure ?? reveals a clear seasonal pattern. Bottom panel (box plots) shows that from January to March, the median LWP remains around  $10 \text{ g m}^{-2}$  with high variability. This changeability shows how different liquid clouds can be at these months, especially in April, when a notable increase in the median LWP occurs. It is expected since these are the months with more occurrence of this type of clouds. Top panel shows the presence of clouds at 3 different levels with LWC varying from 100 to  $200 \text{ mg m}^{-2}$  at 3 different levels. In addition, just a few liquid clouds in June contribute to a very high LWC, ranging from  $100\text{-}500 \text{ mg m}^{-2}$ , exhibiting considerable variability. However, in July, there is a marked decline in the median LWP, indicating a reduction in these clouds. From August to December, the median LWP shows a gradual increase, peaking in



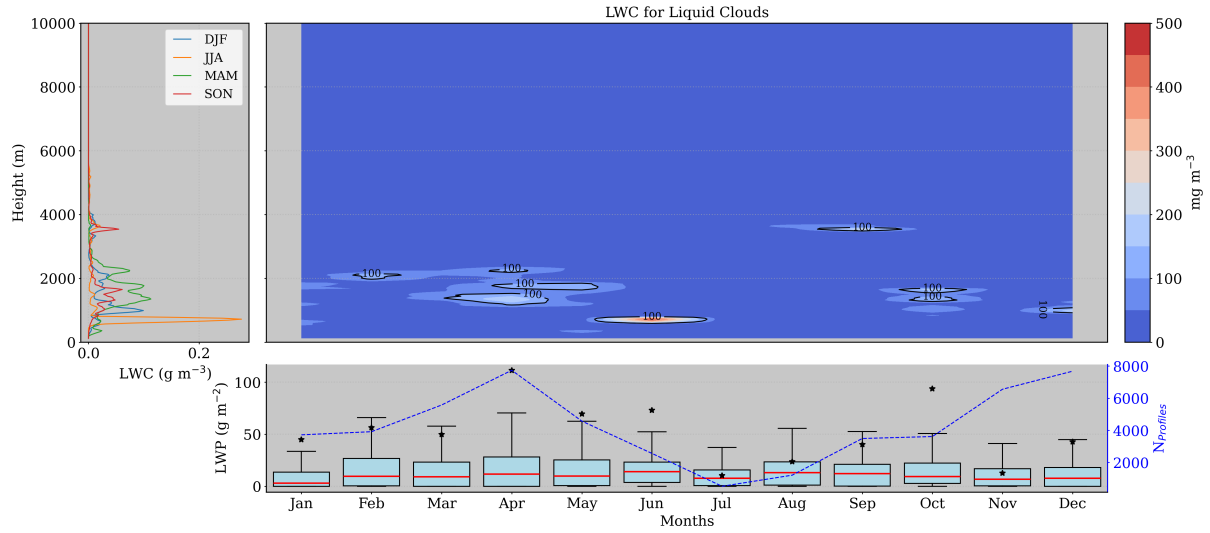
**Figure 3.** Seasonal evaluation of cloud thickness, base and top. The legend above indicates the color correspondent to each cloud type. This figure shows a dark gray box-plot with a white median inside an estimated density distribution. This distribution is displayed in relation to the middle of the box-plot in the right and the left direction.

October, with variability particularly pronounced in transition months like September and October. Additionally, the number of profiles ( $N_{profiles}$ ) varies throughout the year, peaking in April and December, which correspond to increased observation frequency.

Seasonal variations are evident in the LWC profiles by season. In winter (DJF: December, January, February), the LWC exhibits higher values near the surface and up to approximately 2000 meters, indicating a greater presence of liquid water in clouds. During spring (MAM: March, April, May), there is a significant increase in LWC, especially in April, suggesting enhanced cloud liquid water content in this season. Summer (JJA: June, July, August) is characterized by relatively lower LWC values, with minimal liquid water content compared to other seasons. In autumn (SON: September, October, November), the LWC begins to increase again, with values higher than in summer but lower than in spring.

95 The contour plot of LWC further underscores these seasonal patterns, highlighting the vertical distribution of LWC with a color gradient indicating its magnitude. Higher LWC values are observed during spring (April) and early winter (December), which aligns with the trends seen in the LWP box plot and seasonal profiles. The plot shows significant cloud formation at altitudes of 1000 to 2000 meters, with LWC values ranging from 100 to 500  $\text{mg m}^{-3}$  during peak months.

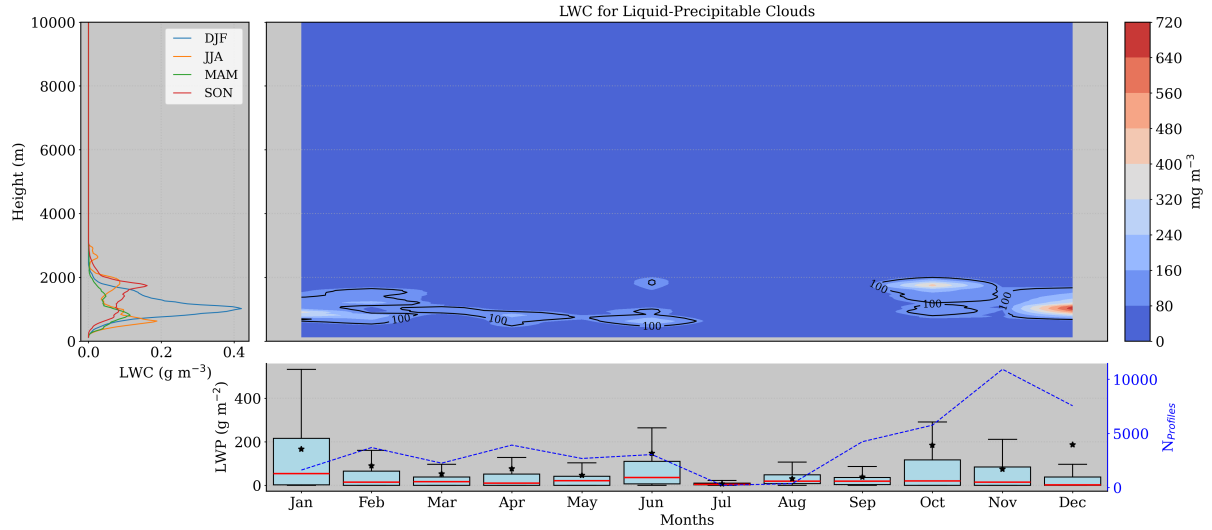
In summary, spring (MAM) and early winter (DJF) are characterized by higher LWC and LWP, indicating increased cloud liquid water content during these periods. Conversely, summer (JJA) shows the lowest LWC and LWP, reflecting reduced liquid cloud presence. Autumn (SON) demonstrates a gradual increase in LWC and LWP as the season progresses. These findings provide valuable insights into the temporal dynamics of cloud liquid water content, emphasizing periods of heightened and diminished cloud water presence.



**Figure 4.** Seasonal and monthly average of liquid water content (LWC) and liquid water path (LWP) for liquid clouds. Left panel shows the seasonal mean profiles of LWC. Right panel shows the monthly mean profiles of LWC and monthly box-plot of LWP. For the boxplots, medians are denoted by red lines and the average by black stars.

Similar seasonality was observed for precipitable liquid clouds, depicted in Figure 5. Nevertheless, LWC are higher due to the presence of drizzle, and peaks are more sharply defined around 1 km during fall, winter and spring. Also, winter achieved higher values, twice as large as in fall and spring ( $0.2 \text{ g m}^{-3}$ ), reaching  $0.7 \text{ g m}^{-3}$ . In summer, values are relatively smaller, with peak of  $0.1 \text{ g m}^{-3}$  around 2 km. Right panel shows that monthly values still fluctuate through several orders of magnitude, and larger values are not well vertically distributed in summer as observed for the only liquid case. They still kept within thin layers, although they reached at slight higher levels. This is in agreement with vertical position of cloud base and top height (see Figure 3), which does not showed a significant vertical variability in summer. In terms of LWP, no seasonality was observed due to the presence of rain droplets, which highly increased its water content in all months. Note that LWP were not separated between cloud and rain, it is the integrated value reported by the microwave radiometer. As a result, slight discrepancy between

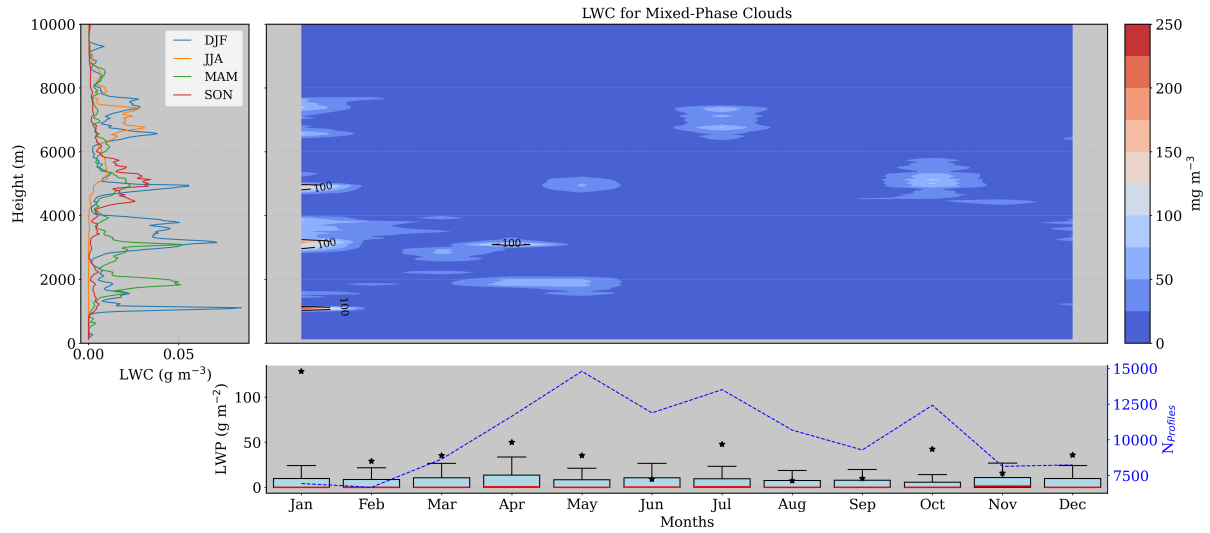
the mean and the median was observed, and its values are roughly 10 - 20 times higher in comparison with the liquid case, with 75% of data below  $250 \text{ g m}^{-2}$ . Also, October presented a significant variation of LWP in relation to other months, with more than  $1000 \text{ g m}^{-2}$  inside the 75<sup>th</sup> percentile. Thus, this month is marked by stronger rains from this particular cloud.



**Figure 5.** Same as Figure 4, but for precipitable liquid clouds.

Mixed-phase clouds exhibited marginally greater occurrence than liquid clouds (Figure 1). Typically, they co-occur and can be comparable in terms of water content. Seasonal average profiles of LWC for mixed-phase clouds in the left panel of Figure 6 shows a similar vertical distribution of LWC between summer and fall (warm regimes), and between winter and spring (cold regimes). Below 4 km, cold regimes have higher average LWC, and similar values with warm regimes above it. In general, these profiles have just smaller differences, and average values are always less than  $0.05 \text{ g m}^{-3}$ . This shows that liquid clouds, besides being thinner, have more water than mixed-phase clouds, suggesting that mixed-phase clouds are mainly composed by ice hydrometeors. Next, monthly mean time evolution on the right panel shows a spreader vertical distribution of LWC, which becomes thinner and higher in summer. This suggests that liquid layers inside mixed phase clouds become thinner in summer, and the ice content is still high enough to keep its thickness around the seasons, as observed in figure 3. Additionally, the right-bottom panel of Figure 6 shows greater occurrence of large LWP values at the end of winter and during spring. Overall, 84% of the analyzed mixed-phase cloud profiles have LWP below  $100 \text{ g m}^{-2}$ , emphasizing the point these clouds have less water than liquid clouds, which has 78% as already discussed.

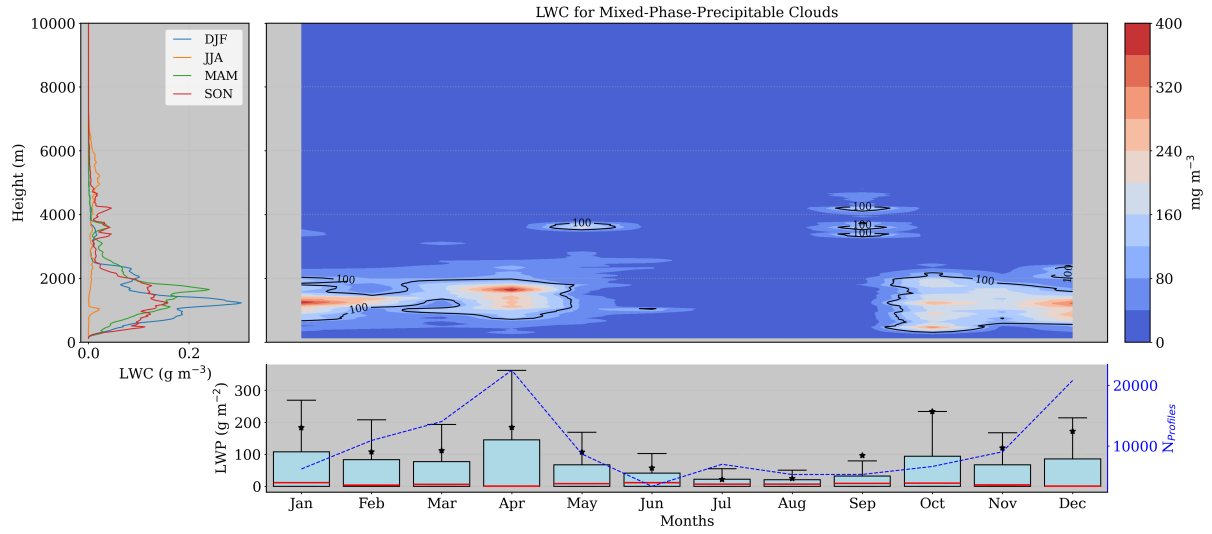
Left panel of Figure 8 reveals that precipitable mixed-phase clouds present similar average profiles up to 4 km in fall, winter and spring. During summer, average LWC within this layer are more than twice smaller in comparison with other seasons. On average, these clouds still have less water content than precipitable liquid clouds below 2 km. This evidence, combined with the observation that liquid layers are notably narrower than mixed-phase layers, suggests that liquid hydrometeors represent a mere fraction of mixed-phase clouds. This observation has more weight here since precipitable mixed-phase clouds have much more



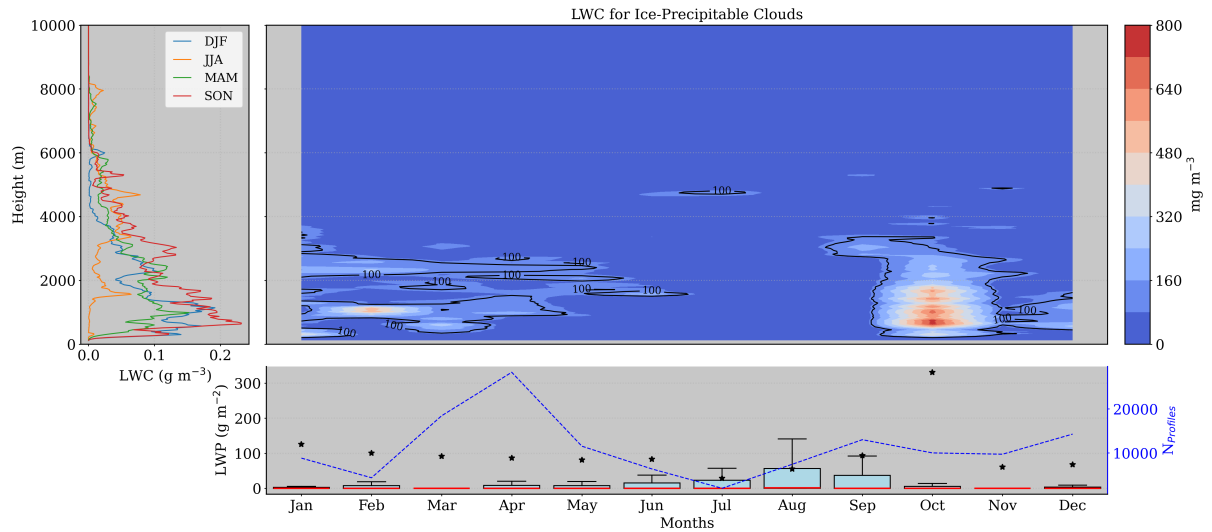
**Figure 6.** Same as Figure 4, but for mixed-phase clouds.

frequency of occurrence. In addition, the blue axis in the bottom-right panel of Figure 8 shows a relevant higher number of cloud profiles, with a sudden decrease in summer. Monthly evolution of LWC clearly indicates a concentration of LWC below 4 km, except in summer, when these clouds have a higher cloud base. In general, the observed mean values are higher than only mixed-phase clouds due to the elevated presence of drizzle. In this case, LWC has a narrower vertical distribution and is closer to the ground. Therefore, they will also have much higher LWP than mixed-phase clouds, as depicted in the right-bottom panel. In terms of its monthly median, no clear seasonal pattern is identified. However, as in the precipitable liquid case, a strong variability of LWP was found in October. This time, we can assume with more statistical confidence that rain events in October are more intense. Although precipitation is more frequent during spring. Moreover, precipitable mixed-phase clouds exhibit 70% of LWP below  $250 \text{ g m}^{-2}$ ; in other words, they can attain higher values than precipitable liquid clouds.

Overall, a notable reduction on LWC was observed in summer. Also, values were found at high levels in the atmosphere in this season, in accordance with the cloud base height pattern. These differences were more readily observed by mixed-phase and precipitable mixed-phase clouds. On the other hand, no seasonal pattern of LWP was clearly identified for all clouds, owing to the high month variability of this variable. To conclude, no clear seasonal pattern was identified for clouds in relation to monthly median LWP. However, liquid clouds have greater LWP than mixed-phase clouds. Once mixed-phase clouds are considerably thicker than liquids, the majority of their composition comprises ice hydrometeors, which will be confirmed in the next section. Moreover, for precipitation cases, the liquid water content increases due the presence of drizzle, and LWP can reach values approximately 10 times greater. In addition, precipitable mixed-phase clouds can achieve higher values of LWP than precipitable liquid clouds.



**Figure 7.** Same as Figure 4, but for precipitable mixed-phase clouds.

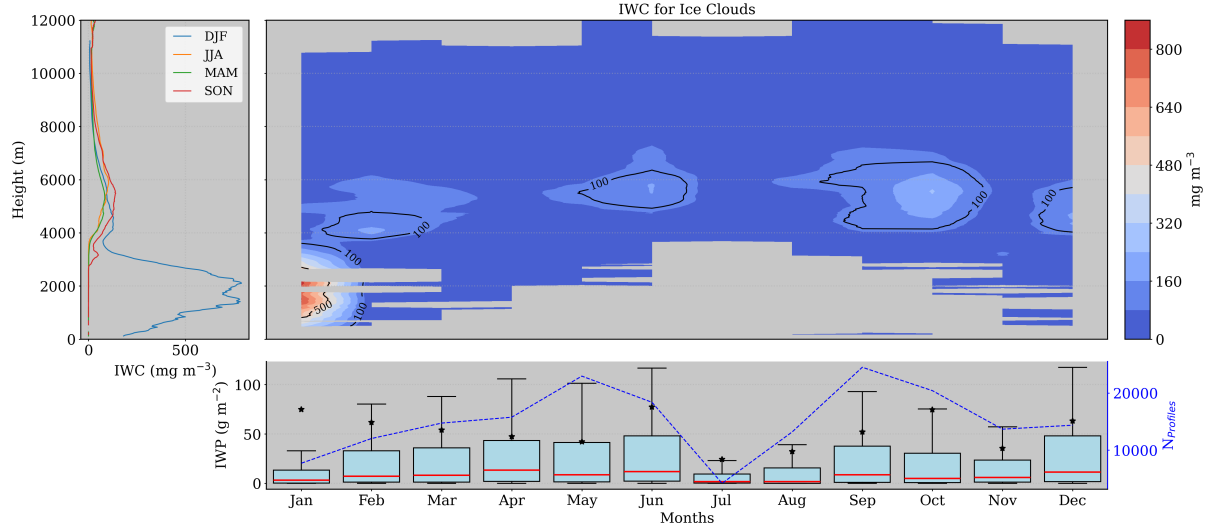


**Figure 8.** Same as Figure 4, but for precipitable ice clouds

#### 1.4 Ice water content and ice water path

Figure 9 displays the average IWC content profiles for each month and seasons, and monthly box-plots of IWP. An evaluation of seasonal profiles on the left panel of this figure denoted a markable presence of IWC at 2 km in winter, with a peak of  $0.2 \text{ g m}^{-3}$ , while at the same level, no IWC was observed during other seasons. Therefore, despite that Figure 3 shows almost no ice cloud base at this level during winter, the presence of just a few clouds with high ice water content are affecting its average.

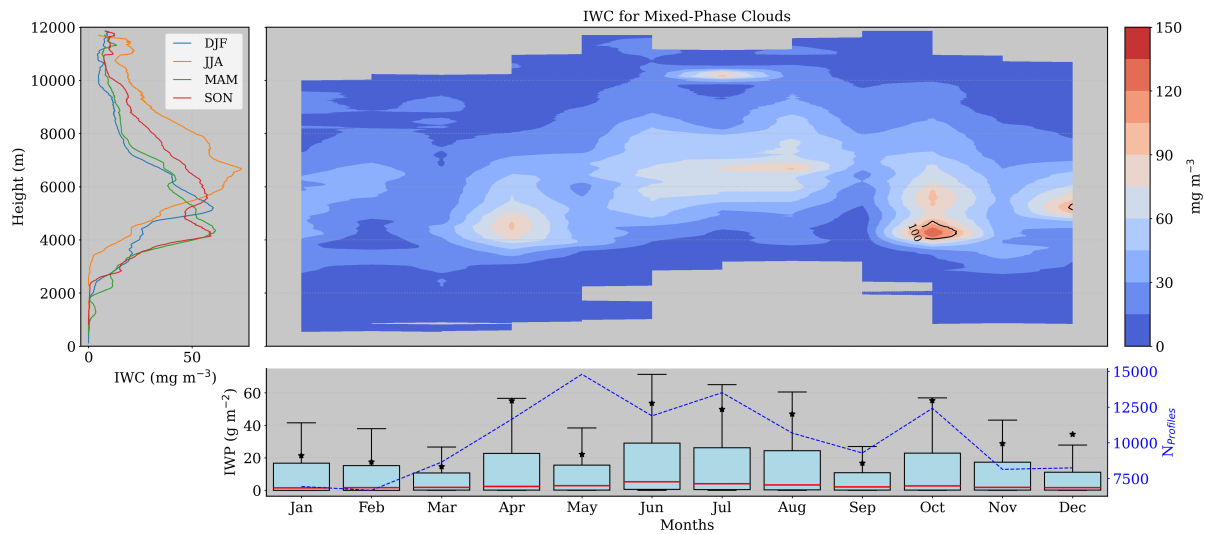
In fact, monthly profiles in the right-top panel reveal that this ice cloud occurs in January. This can be seen by the red tone centered exactly at 2 km in this month. During spring, the peak is around  $0.12 \text{ g m}^{-3}$  at 4.5 km, and above this level its average profile coincides with the winter profile. A similar feature is observed for warm regimes, average profiles of fall and summer coincide above 7 km. Below this level, fall has a peak of  $0.2 \text{ g m}^{-3}$  for about 5 km and summer has a peak of  $0.1 \text{ g m}^{-3}$  around 6 km. In general, ice cloud base heights are usually above 6 km (refers to Figure 3). Hence, relevant values of IWC for these clouds are within 6 and 10 km, in addition, there is no IWC below 2 km, except in January. Concerning the IWP, the median are relatively smaller than the mean, showing a right-skewed distribution with outliers, which are affecting their monthly mean. In terms of medians, no clear seasonal pattern was found. However, larger variability was observed during spring and beginning of fall. In fall (Sep-Nov), ice cloud bases are higher, and average IWP also experienced a maximum of about  $100 \text{ g m}^{-2}$  in September. Regarding the entire dataset, 90% of the IWP are smaller than  $100 \text{ g m}^{-2}$ .



**Figure 9.** Seasonal and monthly average of ice water content (LWC) and ice water path (LWP) for ice clouds. Left panel shows the seasonal mean profiles of IWC. Right panel shows the monthly mean profiles of IWC and monthly box-plot of IWP. For the boxplots, medians are denoted by red lines and the average by black stars.

Mixed-phase clouds exhibited a similar mean profile of IWC during summer and fall as can be seen in the left panel of Figure 10. The only significant difference was during the summer, when mixed-phase clouds do not have a peak around 2 km as observed for ice clouds. Although seasonal profiles suggest mixed-phase clouds have almost the same amount of IWC of ice clouds, it is worth noting that cloud thickness for ice clouds is smaller than for mixed phase clouds. It was observed that mixed phase clouds are almost 1 km thicker than ice clouds. Thus, IWC will be greater, which is the integrated IWC, will be greater as can be seen by the box-plots in the right-bottom panel. This panel shows that IWC easily surpasses  $50 \text{ g m}^{-2}$ , which was not common for ice clouds. In addition, there is less discrepancy between monthly mean and median than for the ice case.

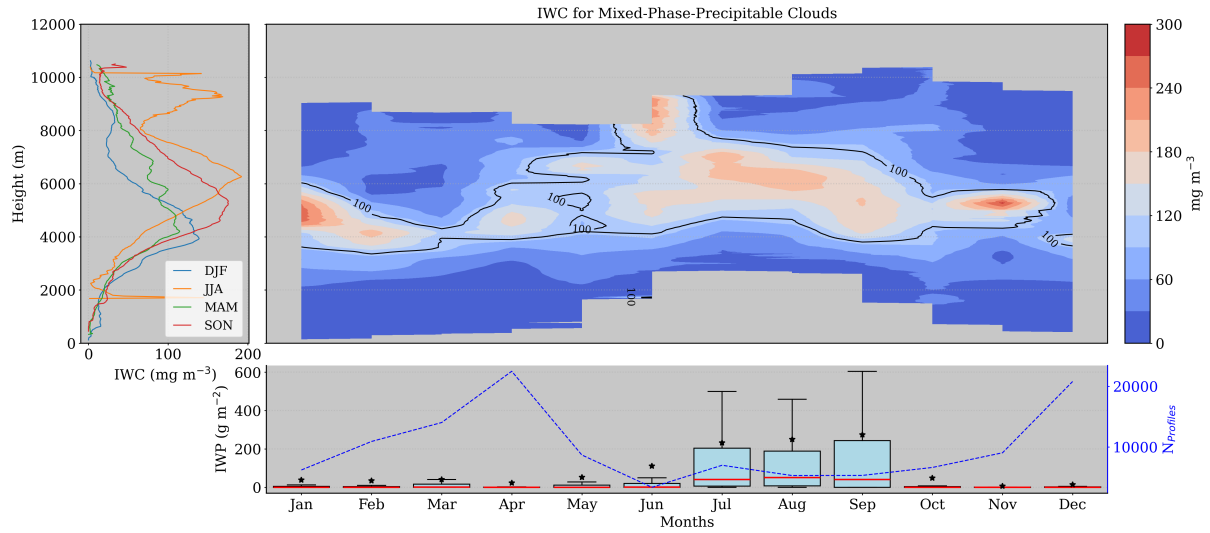
Also, 74% of all IWP for mixed-phase clouds are below  $100 \text{ g m}^{-2}$ , showing that these clouds have more ice content than ice clouds.



**Figure 10.** Same as Figure 9, but for mixed-phase clouds.

For precipitable mixed-phase clouds, winter and spring still have a similar mean profile of IWC (see Figure 11), with peaks around  $0.2 \text{ g m}^{-2}$  at 4 km. Fall still with a peak around 5 km as observed for the ice and non-precipitable mixed-phase clouds, with a just slightly higher value. The most considerable changes happened in summer, where the average profile reached values higher than the other seasons around 6 km, and kept higher up to 12 km. Within this layer, the right panel indicates larger values during summer (Jun-Aug), in red tone. As a result, IWP in this season is significantly higher in comparison with the rest of the period (see right-bottom panel of Figure 11). Although precipitable clouds have much less frequency of occurrence in summer, this frequency is just 4% smaller than mixed-phase clouds. Also, precipitable mixed-phase clouds exhibited a larger cloud thickness, almost 2 km greater than just mixed-phase clouds in this season. By comparing cloud thickness based on similar cloud frequencies, we can infer that precipitable mixed-phase clouds exhibit significantly greater thickness and substantially larger IWP, particularly during the summer and September. Thus, despite the fact these clouds have extremely low frequency in summer, they demonstrate greater vertical development and contain significantly more ice compared to other periods

To summarize, mixed-phase clouds presented higher IWC than ice clouds. The reason can be attributed to the large thickness of the mixed-phase case compared to the only ice case, which is approximately twice as thicker. Thus, less than half of mixed-phase clouds are made of liquid hydrometers, and the rest are ice hydrometeors. Also, precipitable mixed-phase clouds showed even higher contents of ice, especially in summer. Beside their poor frequency of occurrence in summer, these clouds boast significantly greater thickness compared to mixed-phase clouds, carrying substantially more ice.

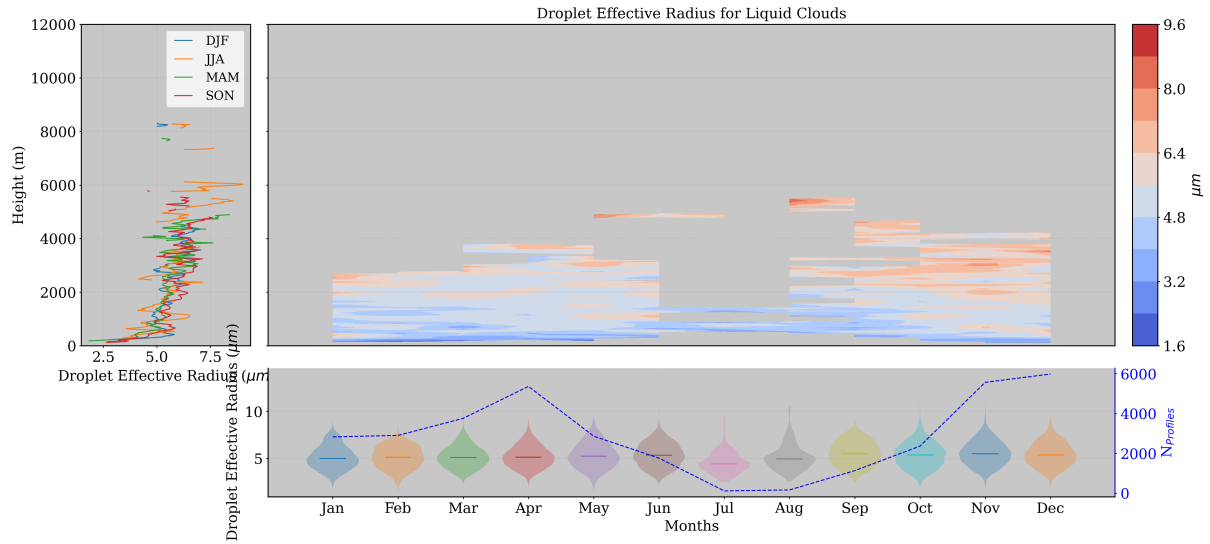


**Figure 11.** Same as Figure 9, but for precipitable mixed-phase clouds.

### 1.5 Droplet effective radius

For liquid clouds, droplet effective radius ( $r_e$ ) were also analyzed by season and monthly. The following analysis was made considering only the median of  $r_e$  and not average values. Left panel of Figure 12 shows the seasonal vertical distribution of the median droplet effective radius where the shaded area represents the 25<sup>th</sup> and 75<sup>th</sup> percentiles. Right-top panel shows the monthly vertical distribution of the median, and the right-bottom panel shows the median and an estimation of the probability density function (p.d.f) of  $r_e$  by month using violin plots. These estimations were made through kernel density method ? and the p.d.f are shown with its reflection with respect to the middle point in the x-axis, with a kind of violin shape. Additionally, this panel has a second y axis with the number of profiles at each month to compute statistics. First, no seasonal difference was observed for profiles on the left panel of Figure 12. Second, monthly profiles on the right-top panel show that large droplets are found above 4 km, in yellow and red tones. However, monthly examination  $r_e$  distribution in the bottom panel demonstrates that the medians are about to 5  $\mu\text{m}$ , corresponding to the soft blue tone below 3 km of height, consistent with the LWC distribution in Figure 4.

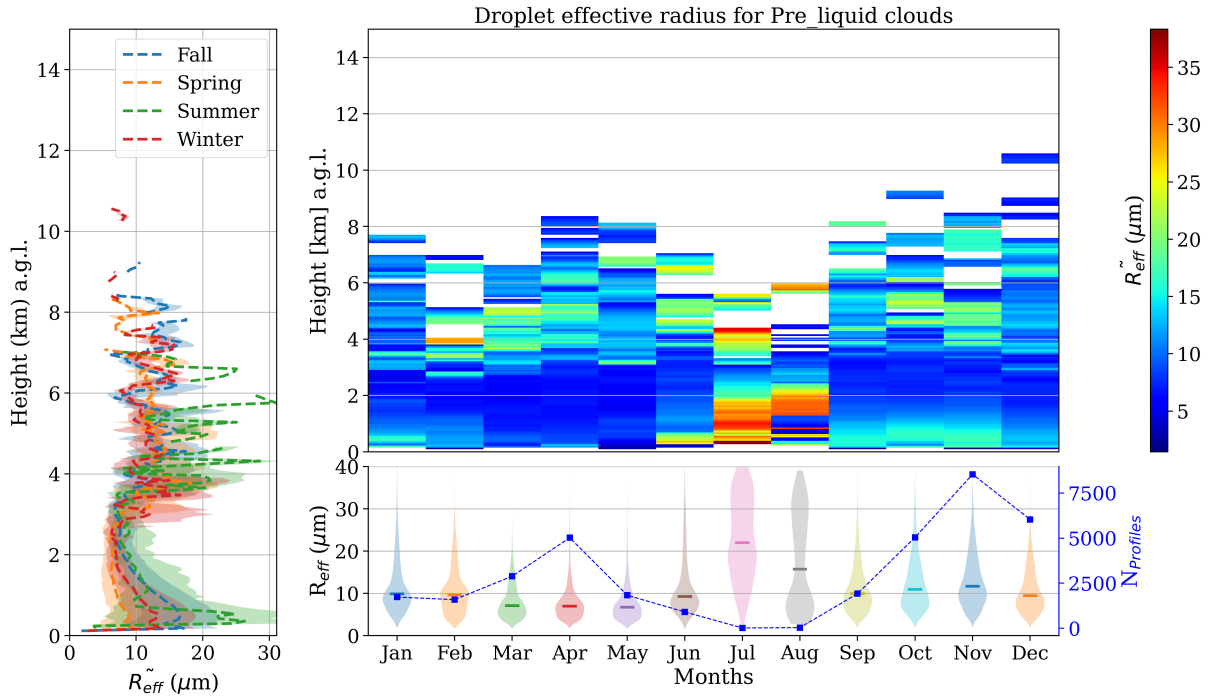
Next, precipitation liquid clouds have a slight difference on seasonal profiles below 2 km, as can be seen on the left panel of Figure 13. Below this level, spring has the lowest values, followed by winter, fall and summer. However, in this case, there are almost no profiles with precipitable liquid clouds in summer. This can be seen by the blue line in the right-bottom panel. Thus it is challenging to evaluate seasonal differences on liquid droplets for these clouds which have almost zero frequency of occurrence in summer. On the other hand, it is clear that their liquid droplets are greater than for liquid clouds. This can be associated with the presence of drizzle, as discussed in terms of LWC. In this case, the median droplet effective radius ranges around 8 and 9  $\mu\text{m}$ , except in summer, where droplets are greater than 15  $\mu\text{m}$ .



**Figure 12.** Seasonal and monthly statistics on droplet effective radius ( $r_e$ ) distribution for liquid clouds. Left panel shows the seasonal median profiles of droplet effective radius, where shaded area denote the 25<sup>th</sup> and 75<sup>th</sup> percentile. Top-right panel shows the monthly profiles of the median  $r_e$ , while the bottom-panel displays the size distribution of  $r_e$  by months.

Concerning mixed-phase clouds (Figure 14), no seasonal differences were observed. Nonetheless, Left panel of Figure 14 shows that droplets of mixed-phase clouds are systematically larger than within liquid clouds, roughly twice the size. In addition, larger cloud droplets are found between 3 and 5 km of height. This is also observed by the monthly median profiles in the right-top panel, by the yellow and red tones. Same panel also indicates that greater droplets sizes are found in summer, despite the fact there are small amounts of LWP in this season as observed in Figure 6. Even though an evaluation of droplet size distribution on the right-bottom panel of Figure 14 shows an stable median around 10  $\mu\text{m}$  through the months, droplets can vary between 13 - 22  $\mu\text{m}$  in the neighbourhood of 4.5 km of height in summer. However, values are typically much smaller than this, as can be seen by the monthly p.d.f in the right-bottom panel. They show a mode below the median during the entire year, revealing that the most frequent droplet sizes are even smaller than 15  $\mu\text{m}$ .

Differently, precipitable mixed-phase clouds do not exhibit vertical variation of droplet size. Their median values were kept almost the same along the vertical by each median profile of Figure 15. Also, summer showed a slightly higher median profile of  $r_e$  than other seasons, roughly 10  $\mu\text{m}$  greater than winter, between 3 and 6 km. Differences below this layer are quite challenging to analyze since these clouds have a cloud base about 3 km in summer, as already observed in Figure 3. Thus, besides these clouds having almost no occurrence in summer, there is almost no cloud base below this level. Therefore, a seasonal comparison of  $r_e$  should be done above this level, which still has larger  $r_e$ , but differences are less discrepant. An evaluation of the monthly distributions on the right-bottom panel also attribute less weight to these low clouds during summer. As a consequence, a difference around 5  $\mu\text{m}$  was also observed between winter and summer medians. In summer, the median is varying around 20  $\mu\text{m}$ , and during the winter, the median is about 15  $\mu\text{m}$ . In addition, during summer it is possible to observe



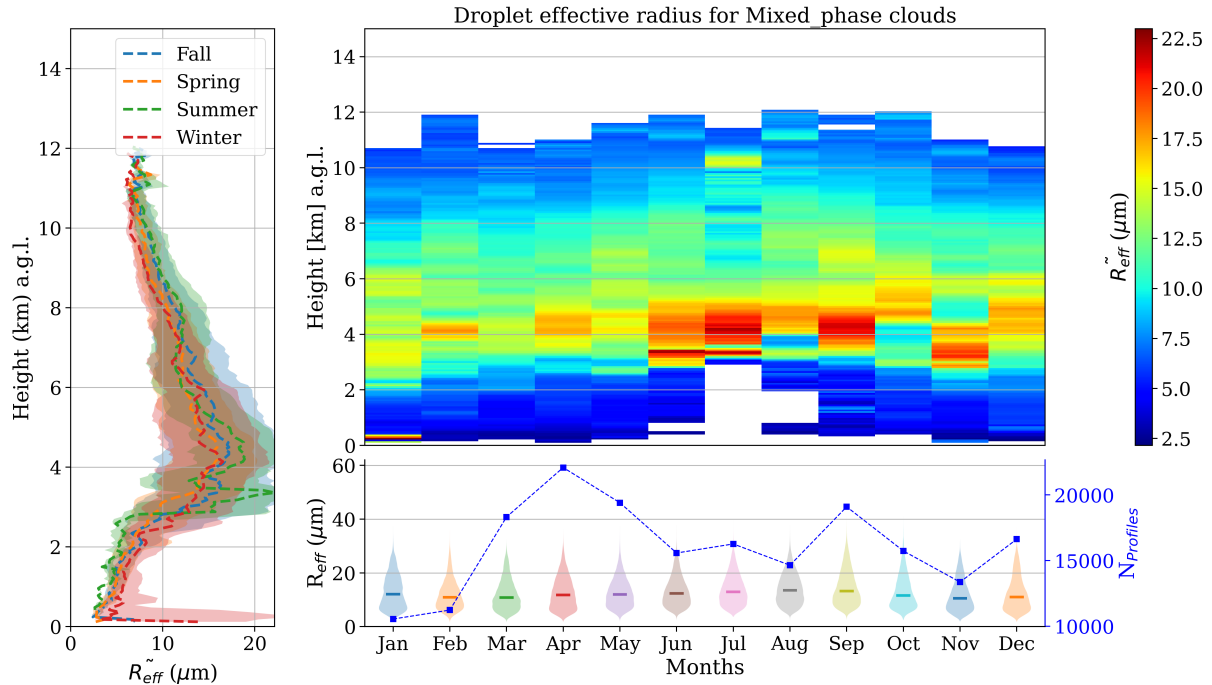
**Figure 13.** Same as figure 12, but for precipitable liquid clouds.

that the position of the mode, which is located below the median for the rest of the period, starts to move until stays above the median in August.

230 In general, no strong seasonal differences were observed for liquid and precipitable liquid clouds. Liquid clouds have  $r_e$  varying around 5  $\mu\text{m}$ , while precipitable clouds achieved higher values due to the presence of drizzle droplets, varying between 10  $\mu\text{m}$ . Mixed-phase clouds showed consistently higher median  $r_e$  compared to liquid clouds, where larger values occurred around 3 and 5 km of height, especially during summer. Since just a few fraction of these clouds have larger  $r_e$ , and statistics with all monthly data showed only small variation of the month medians, which stayed around 15  $\mu\text{m}$ . In relation to precipitable  
 235 mixed-phase clouds, larger differences observed in summer occurred due to the presence of a few low clouds, which were higher  $r_e$ . However, these clouds do not have enough occurrence to compare with the other months, which have considerably strong statistics.

## 1.6 Ice effective radius

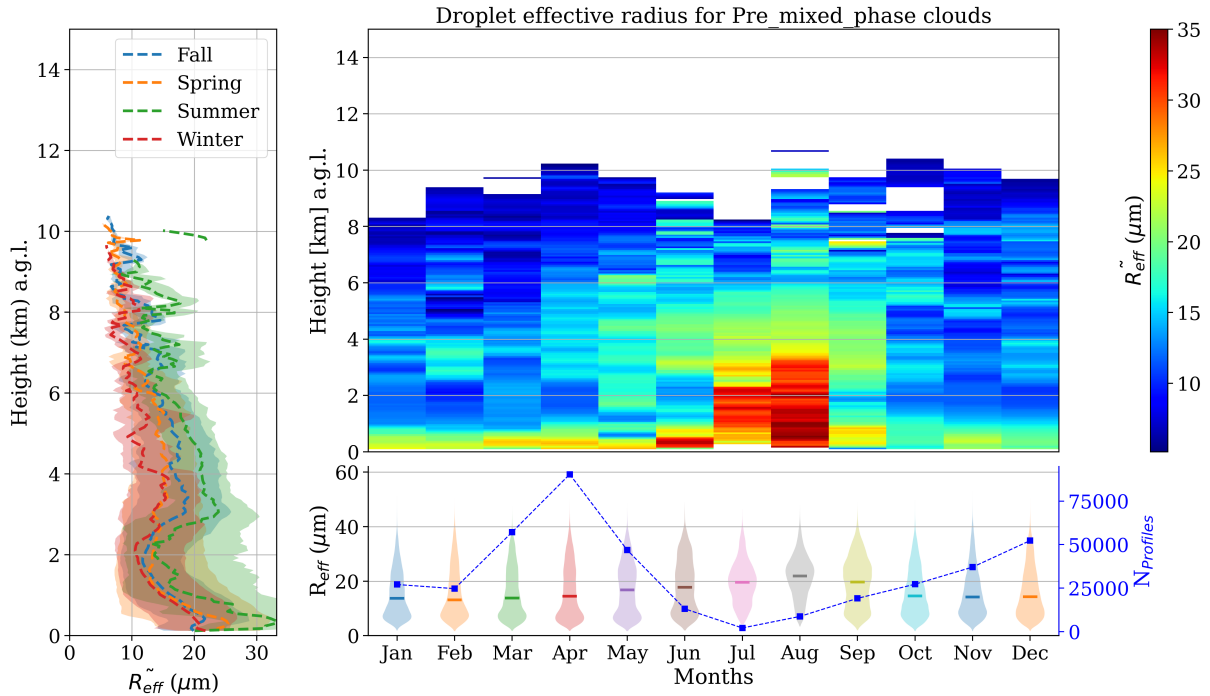
Ice effective radius for ice clouds are depicted in Figure 16, and is organized as the previous droplet sizes analysis. Left  
 240 panel shows a slight seasonal difference above 7 km, where summer assumed larger median  $r_e$ . Also, larger values of ice  $r_e$  is observed in summer by the yellow tone bellow 4 km in the right-top panel, indicating a radius about 60  $\mu\text{m}$ . However, this clouds have a median cloud base around 7 km in summer, in addition to a 25<sup>th</sup> percentile of 5 km. This shows that there is a



**Figure 14.** Same as figure 12, but for mixed-phase clouds.

negligible occurrence of ice clouds below 4 km. Therefore,  $r_e$  for these clouds are typically ranging between 30 and 50  $\mu\text{m}$ , where lower values are more frequently. This can be also seen in the monthly p.d.fs on the right-bottom panel. Throughout the observation period, the median remained close to 30  $\mu\text{m}$ , and increased in summer due to the small presence of low ice clouds. During this season, median  $r_e$  are close to 50  $\mu\text{m}$ . It is worth to note the importance of analysing the median instead of the mean in these cases. Clearly, outliers close to surface, which does not represent a real situation for ice clouds will affect the average in May, August, September and October. The monthly distribution of  $r_e$  also shows the presence of a mode below the median for almost all season, except for April and summer. As a result, common  $r_e$  can be even smaller than 30  $\mu\text{m}$ .

For mixed-phase clouds, the left panel of Figure 17 also shows a slight difference between profiles above 6 km. In summer and fall, median values are slightly larger than other seasons. The reason can be associated with the vertical location of clouds at each season, as cloud bases in summer and fall are around 5 km, and one kilometer lower in winter and spring. Hence, high mixed-phase clouds in summer and fall are exhibiting large  $r_e$ . Seasonal differences below 3 km are quite challenging to evaluate due to the negligible presence of these clouds. As a reference, the 25<sup>th</sup> percentile of cloud base height in summer is 4.6 km, denoting a small presence of clouds below this level. However, regarding the entire dataset, the distributions in the right-bottom panel presented slightly larger values in summer and fall, reaching at 50  $\mu\text{m}$  in July, August and September. In general, median  $r_e$  are just a little smaller, with high variability. Different from the previous case, the median and the mode almost coincide for all the months.



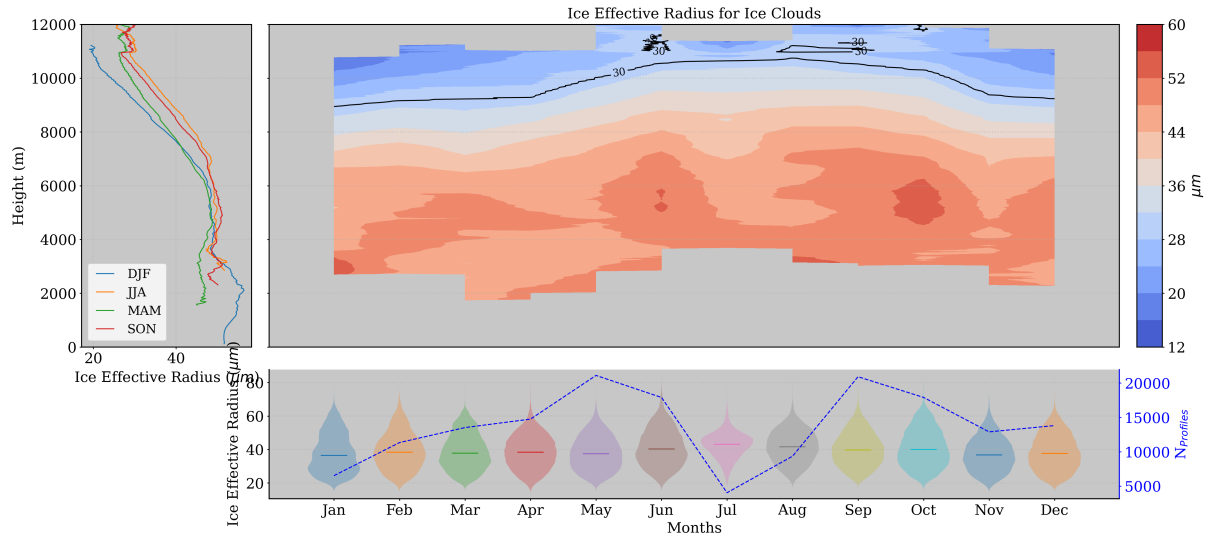
**Figure 15.** Same as figure 12, but for precipitable mixed-phase clouds.

Lastly, precipitable mixed-phase clouds experienced the greater seasonal difference between median profiles. Above five  
 260 kilometers, summer has the largest median values, followed by fall and then winter and spring with the same values. Differences  
 above 7 km cannot be analyzed since cloud tops are frequently not observed at this level, with the 75<sup>th</sup> percentile at 6 and  
 4.5 km in fall and spring, respectively. Probably, the differences observed above 12 km are due to a bad classification of cloud  
 layers. Same reason can be causing values about 70  $\mu m$  at 10 km observed in the right-top panel of Figure 17, in July. In  
 addition, this month has a poor number of profiles, denoting just minor cases that can be contributing to these high values. In  
 265 general, the statistical median of the monthly p.d.fs is stable in 50  $\mu m$ , with a narrower distribution.

In summary, ice clouds have median  $r_e$  ranging between 30 and 50  $\mu m$ , where values can be larger at more elevated levels.  
 The second is commonly found during summer, and the first during the rest of the period. For mixed-phase and precipitable  
 mixed-phase clouds, the median  $r_e$  are about 50  $\mu m$ . In fact, mixed-phase clouds have slightly smaller values than 50  $\mu m$   
 excluding summer, and a broader distribution. However, values are quite similar with the precipitable case.

## 270 Appendix A: Cloud assessment by time

This method have been used by recent studies such as Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al..  
 Each study has adopted some different criteria to classify single layer clouds, but they all classify cloud layers by profile. Here,

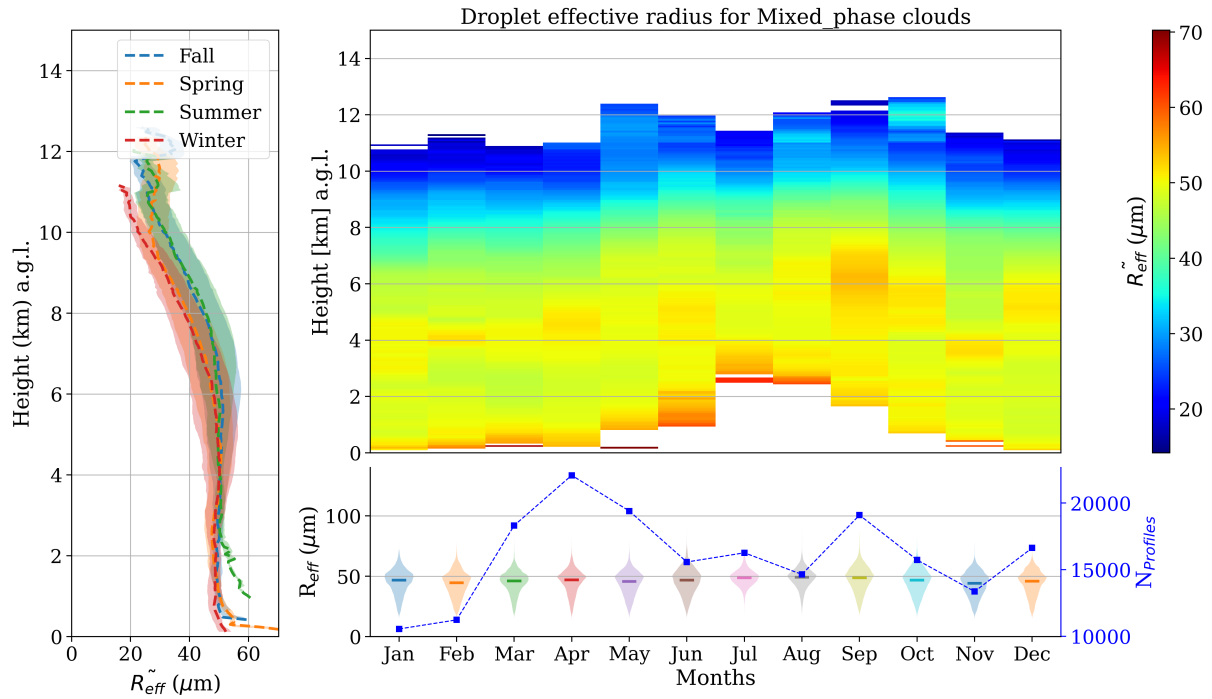


**Figure 16.** Seasonal and monthly statistics on ice effective radius ( $r_e$ ) distribution for ice clouds. Left panel shows the seasonal median profiles of ice effective radius, where shaded areas denote the 25<sup>th</sup> and 75<sup>th</sup> percentile. Top-right panel shows the monthly profiles of the median  $r_e$ , while the bottom-panel displays the size distribution of  $r_e$  by months.

we adopted some criteria based on these studies and employed the following method: cloud layers in atmospheric column were classified through pixel sequence inquiry. These pixels sequences were divided in the following 5 groups according to Răzvan Pîrloagă et al.: liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only "ice" pixel sequence was encountered. As well as, mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified whether at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Different from previous studies, whether rain pixels are found with more than 20 pixel (256-1022 m) of distance from cloud base, the cloud are not classified as precipitable, assuming that rain droplets are not falling from these cloud layer. In all cases, for being considered a cloud layer, a minimum sequence of 3 cloud pixels (38-153 m) (?) must be satisfied, in addition to a minimum sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Aerosol & insect" to separate multi-layers. Therefore, free hydrometeor layers in the neighbourhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on ??, but these values are not currently well defined, their determination has been guided by empirical experience.

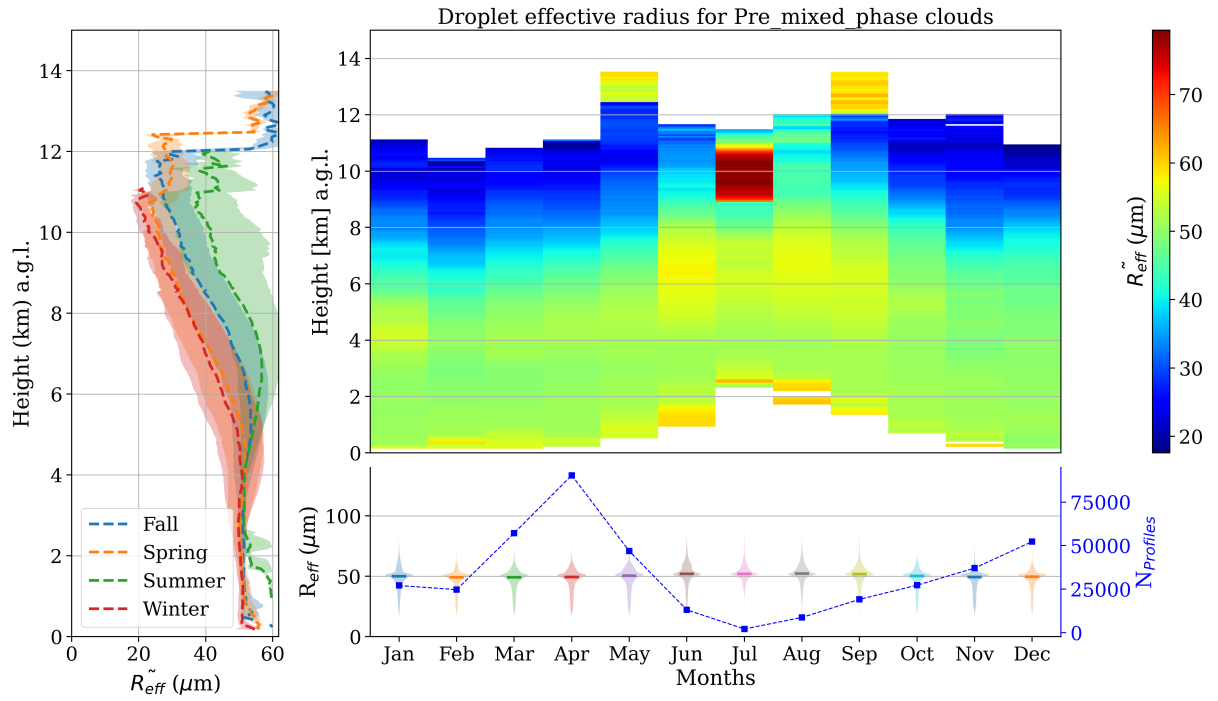
## Appendix B: Cloud dilatation

The dilatation method consist in expand the ones in the cloud mask to connect the closest groups of hydrometeors, which are likely to belong to the same cloud. The dilatation follows the scheme bellow, where hydrometeor pixels are represented in blue.

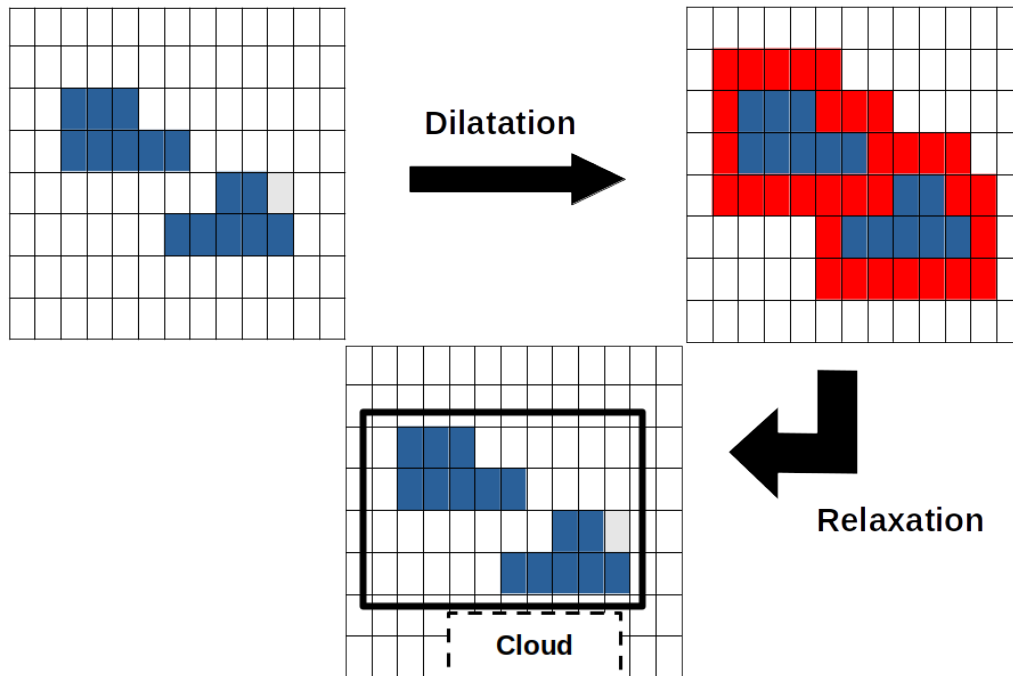


**Figure 17.** Same as figure 16, but for mixed-phase clouds.

Left image represents the original hydrometeor pixels where two groups of clouds were identified. Thus, a dilatation of 1 pixel  
 290 is applied in all directions, represented by the red pixels in the right image. Therefore, the dilatation process will unite these  
 groups with at least 2 pixels of separation distance. Since they are connected, the cloud classification algorithm will save the  
 coordinates of these cloud pixels in just one group (not two, as previously identified).



**Figure 18.** Same as figure 16, but for precipitable mixed-phase clouds.



**Figure B1.** Temporary figure

Next, a contraction process is done to return the coordinates of true hydrometeors to the original position. The contraction will remove the coordinates of fake hydrometeor pixels (pixels in red on the above figure). Afterwards, the classification algorithm can follow its chain and calculate the percentage of each hydrometeor type inside each group to classify different clouds in the cloudnet product.

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